

Min Yang

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RESEARCH INTERESTS

PhD Candidate in Biomedical Engineering focusing on **AI, quantum computing** and its applications in the **biomedical** domain, specifically in **metabolism and cancer**.

EDUCATION

Oregon Health and Science University (OHSU), Oregon, USA July 2024 - June 2028
Doctor of Biomedical Engineering, Ph.D. Advisor: **Dr. Chi Zhang** & **Dr. Sha Cao**, GPA: 3.77/4.0

Indiana University Bloomington (IUB), Indiana, USA August 2023 - May 2024
Doctor of Intelligent System Engineering, GPA: 4.0 / 4.0

Xi'an Jiaotong University (XJTU), Xi'an, China August 2020 - June 2023
Master of Engineering in Software Engineering, GPA: 3.49 / 4.0

Xidian University (XDU), Xi'an, China August 2016 - June 2020
Bachelor of Engineering in Software Engineering, GPA: 3.39 / 4.0

PUBLICATIONS

- M. Yang**, X. Guo, L. Jiang. "Multi-Stage Watermarking for Quantum Circuits." in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*.
- M. Yang**, Y. Li, S. Wang, et al. "RITNet: A Rotation Invariant Transformer based Network for Point Cloud Registration." in *2022 IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*.
- Z. Fu, **M. Yang**, C. Chu et al. "QuantumLeak: Stealing Quantum Neural Networks from Cloud-based NISQ Machines" in *2024 International Joint Conference on Neural Networks (IJCNN)*.
- F. Chen, et al. , **M. Yang**. "NISQ Quantum Computing: A Security-Centric Tutorial and Survey [Feature]" in *2024 IEEE Circuits and Systems Magazine*.
- M. Tsuda, et al. , **M. Yang**. "A window trial in metastatic pancreatic ductal adenocarcinoma reveals resistance mechanisms to targeting the KRAS-MEK pathway" in *bioRxiv*, 2025.
- Y. Zhao, J. Yu, **M. Yang**, C. Zhang. "A quantum-enabled atlas-scale resource to study metabolic landscapes and heterogeneity in cancer TME" in *Cancer Research*, 2026.
- A. Li, A. Mao, Q. Wei, **M. Yang**, et al. , "Computational Hardness and Flow-Based Optimization for Neural Connectivity Inference" in *14th International Conference on Intelligent Biology and Medicine (ICIBM)*.
- P. Swaminathan, et al. , **M. Yang**. "Characterizing biological functional changes in cancer tumor microenvironment by integrating histology-based gene imputation with scRNA-seq foundation models" in *Cancer Research*, 2025.

RESEARCH EXPERIENCE(SELECTED)

- AI Scientist for Verifiable Quantum Advantage Discovery** June 2026 – Present
- Developing a **multi-agent AI Scientist** that extracts domain knowledge from scientific literature and autonomously proposes high-value **quantum computing applications**.
 - Developed an end-to-end evaluation pipeline that validates proposed applications by selecting appropriate quantum and classical algorithms, generating experiments, and benchmarking verifiable **quantum advantage**.
- Quantum-enhanced Algorithms for Human Metabolic Analysis** Dec. 2024 – Present
- Developing **hybrid classical-quantum algorithms** to encode metabolic network and transcriptomic data to infer metabolic status.
 - Addressing challenges posed by whole-body and tissue-specific **digital twins** through quantum computing.
- Computational Analysis of in 3D Bioprinted PDAC Models** Oct. 2024 – Present
- Analyzed cell-type composition in 3D bioprinted **PDAC** models across time points and environmental conditions.
 - Characterized dynamic responses of cancer-associated fibroblast subpopulations to microenvironmental changes.
- Quantum Computing Security** August 2023 – May 2024
- Designed a multi-stage watermarking scheme to protect **quantum circuits** from unauthorized replication.
 - Developed QuantumLeak to address noise challenges in extracting **Quantum Neural Network** models from NISQ computers.
 - Identified vulnerabilities and proposed taxonomies of security threats in quantum computing.
- 3D Point Cloud Registration and Semantic Scene Reconstruction** August 2020 – June 2023
- Developed Transformer-based methods for **3D point cloud** registration and semantic mapping.
 - Constructed grid-based semantic scene representations for high-precision road environment understanding.

TECHNICAL SKILLS

- **AI:** PyTorch, RAG, BioAI Agent, AI Scientist, Biosecurity
- **Quantum:** Quantum annealing, QAOA, QUBO, Quantum-inspired algorithms, Quantum machine learning
- **Programming & Data:** Python, R, CUDA, D-Wave Ocean SDK, IBM Qiskit, IonQ, CUDA-Q

SERVICE & HONORS (SELECTED)

- **Invited Talks:** NIH Invited Talk on Quantum Computing, ISMB/ECCB (2025).
- **Peer Review:** Reviewer for QCNC (2025); IJCNN (2024–2025); QCCC (2024); PRCV (2023); ICRA (2023).
- **Awards & Fellowships:** DAC Young Fellowship (2024); Best Presentation Award, DAC Young Fellow Program (2024); First Prize, University of Minnesota 3-Minute Thesis Competition (2022).